

Training Configuration:

Sequencing:

- timesteps: 3
- batch_size: 512

Model Hyperparameters:

- min_units: 8
- max_units: 16
- units_step: 8
- dropout_min: 0.2
- dropout_max: 0.5
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: lstm_hyperparameter_tuning

LSTM Training Report

Report Generated: 2025-04-08_13-52-42

Training Start: 2025-04-08 13:52:09

Training End: 2025-04-08 13:52:20

Training Duration: 11.87 seconds

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.30

Units per Layer:

Layer 1: 128 units

Performance Metrics:

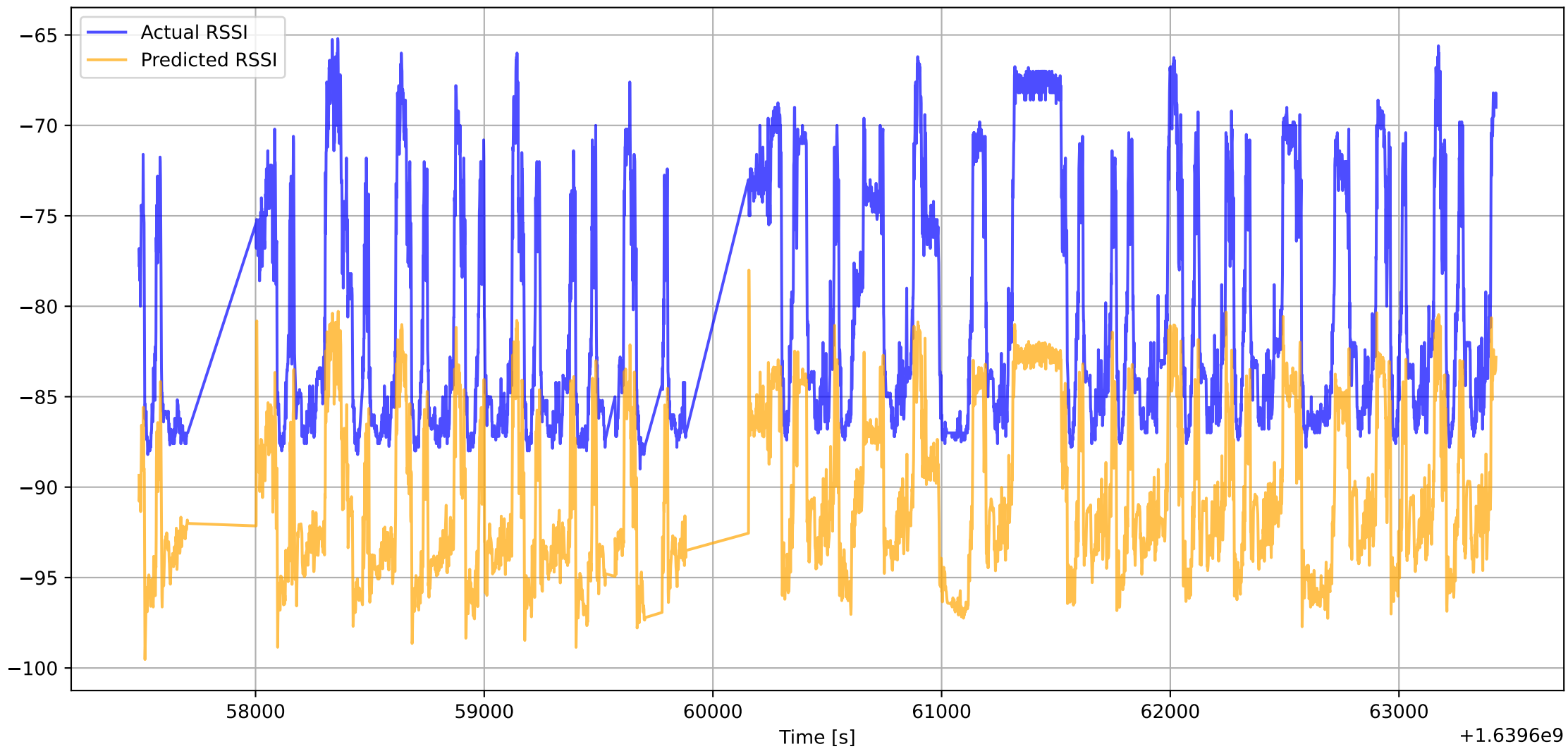
MAE: 10.2935

MSE: 114.5275

RMSE: 10.7018

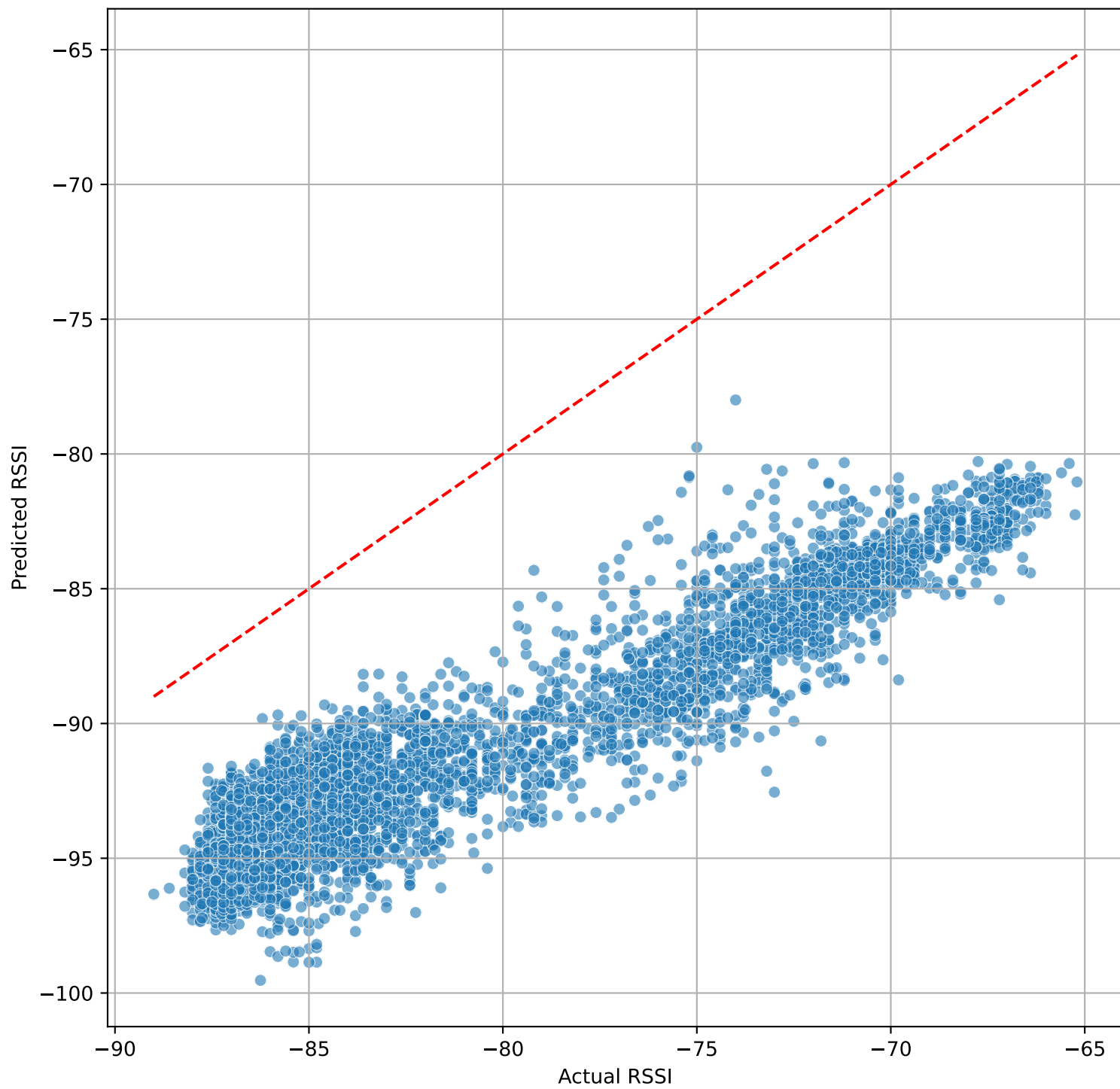
R2 Score: -1.6034

Actual vs Predicted RSSI Over Time

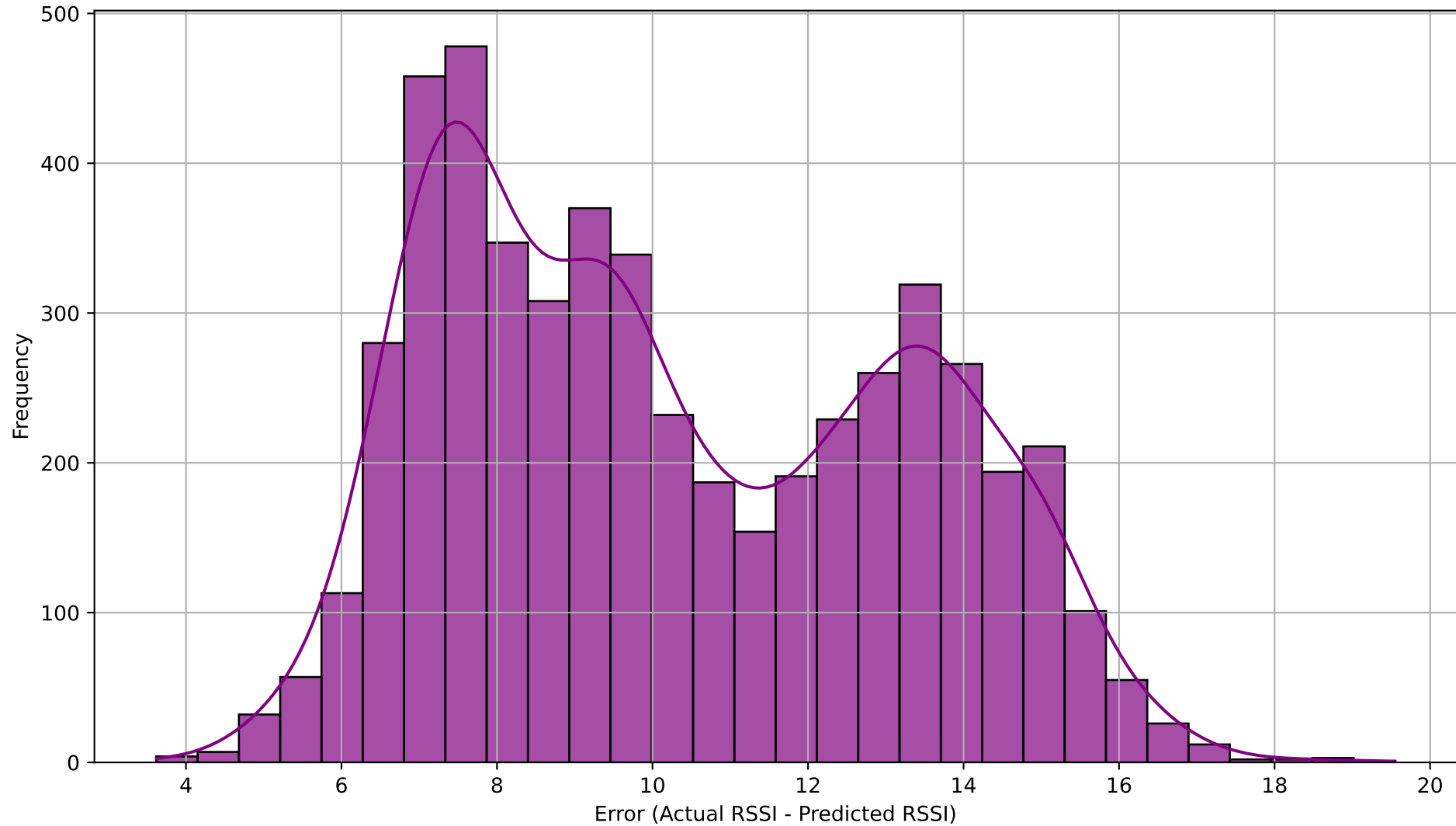


+1.6396e9

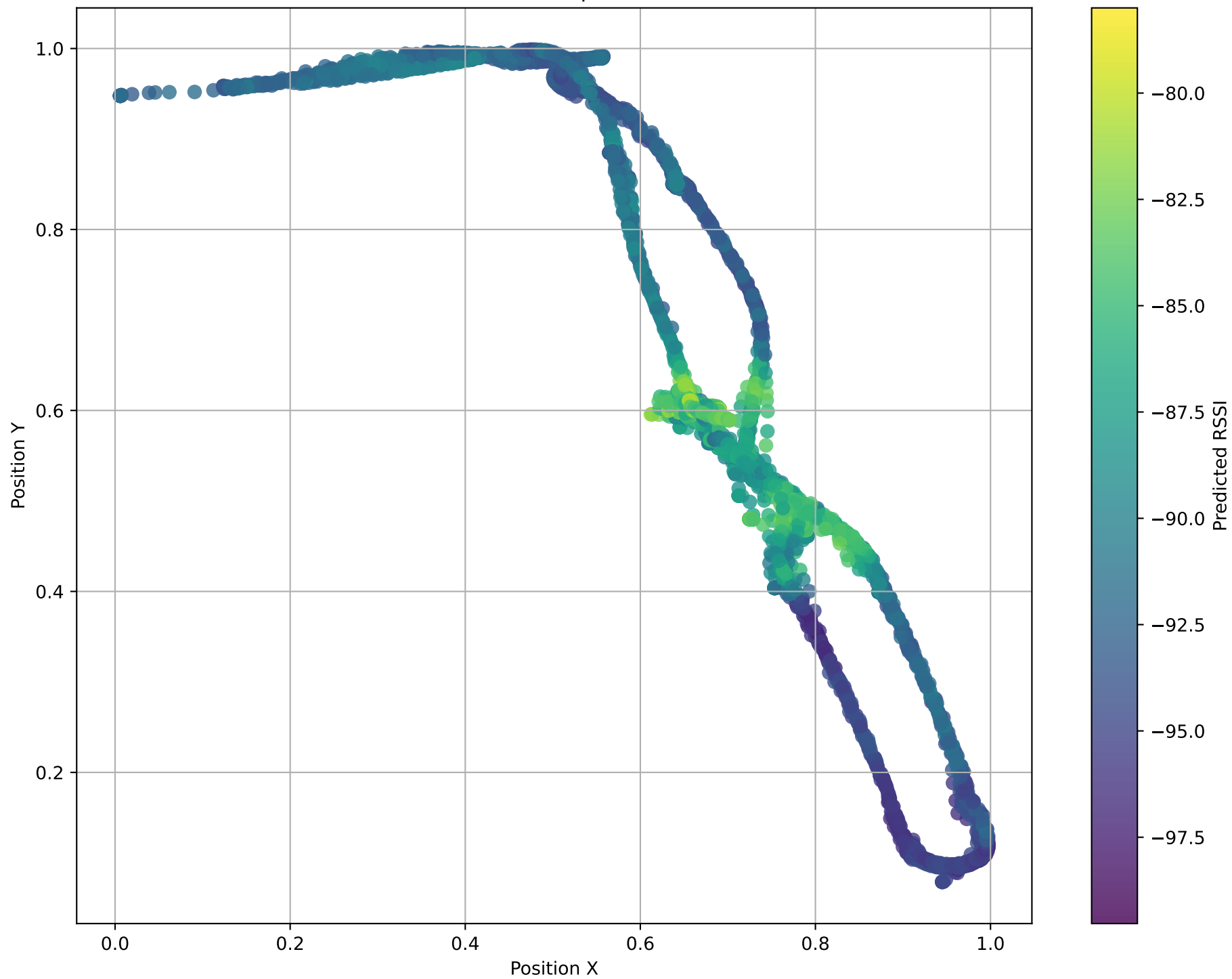
Actual vs Predicted RSSI



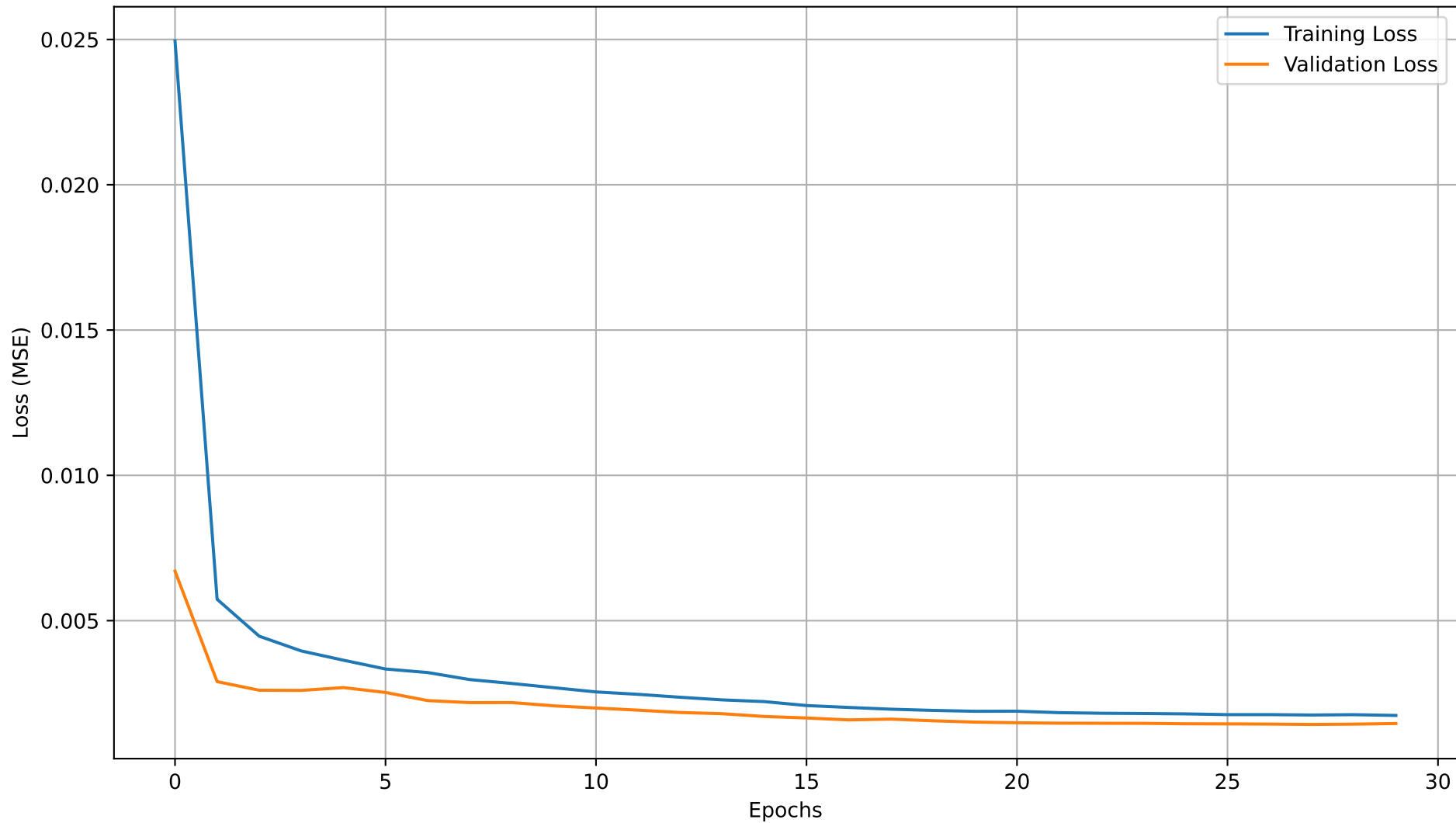
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

